

Solve quadratic equations by factoring (a = 1)

Algebra I | CK-12: Using the Zero-Product Property

Key Idea

To solve $x^2 + bx + c = 0$: get one side to 0, **factor**, then set **each factor equal to 0** (Zero Product Property) and solve.

Key Terms

- **Zero Product Property** — if $AB = 0$ then $A = 0$ or $B = 0$
- **Root / solution** — an x that makes the equation true

Worked steps

Step 1: Move everything to one side so it equals 0.

Step 2: Factor, set each factor = 0, and solve.

Example: $x^2 + 5x + 6 = 0 \Rightarrow (x + 2)(x + 3) = 0 \Rightarrow x = -2, -3$.

You Try

Solve $x^2 - x - 12 = 0$.